

Combinations



1

a 120

b 17280

c 462

d 3876

e 21

f 35

g 10

h $\frac{6}{5}$

2 128

3 16 777 216

4

a 4512

b 17 296

c 21 808

5 211 926

6

a 252

b 210

c 126

Probability with combinations

7

a $\frac{1}{1287}$

b $\frac{25}{1287}$

c $\frac{925}{1287}$

d $\frac{56}{1287}$

8 $\frac{1}{2}$

9 $\frac{253}{9996}$

10 $\frac{12}{35}$

11

a $\frac{66}{195\,755}$

b $\frac{83\,486}{195\,755}$

12

a 15

b 1



13
a 792

b $\frac{1}{12}$

c $\frac{1}{220}$

Applications

14
a $\frac{4}{15}$

b $\frac{22}{65}$

15
a $\frac{91}{204}$

b $\frac{113}{204}$

16
a $\frac{4}{143}$

b $\frac{49}{143}$

17 125 970

18 $\frac{42}{143}$

19 91

20 6006

21 7350

22 27 720

23
a 15

b 63

24 324 632

25 120

26
a 16

b $\frac{1}{16}$



c $\frac{1}{16}$

d $\frac{1}{4}$

27
a 10

b $\frac{2}{5}$

28 20

29
a 165

b 95

30
a 20

b 6

c 30

31
a 126

b 35

c $\frac{5}{18}$

32
a 20

b $\frac{1}{2}$

33
a 150

b 310

34 30

35 126

36 672

37 435

38 128

39
a 325

b $\frac{1}{325}$

40

a 1.01%

b 35.35%



c 2.02%

41

a 5.52%

b 0.01%

c 0.07%

42

a $\frac{429}{5168}$

b $\frac{4739}{5168}$